

YOUTUBE API SERVICES

Compliance Verification Report

Application Name:	Verdent
Application URL:	https://verdent.app
API Product:	YouTube Data API v3
Report Version:	1.0
Prepared Date:	June 2026
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This document is prepared for submission to Google as part of the YouTube API Services compliance review process. It demonstrates how Verdent fetches, processes, and displays data obtained through the YouTube Data API v3, and confirms adherence to all applicable YouTube API Services Terms of Service.

1. Application Overview

Application Name: Verdent

Application URL: <https://verdent.app>

Platform: Web application (macOS desktop client also available)

Verdent is a YouTube channel analytics and keyword research tool designed for individual content creators, digital marketers, and media researchers. The application enables users to search for YouTube videos by keyword, analyze video performance metrics, compare channel growth, and estimate revenue potential based on publicly available engagement data.

The application does not allow end-users to upload, modify, or delete YouTube content. All interactions with the YouTube Data API v3 are strictly read-only. No user authentication via OAuth is performed; the application uses an API key scoped exclusively to read access.

Purpose of YouTube API Usage

- Retrieve video search results based on user-supplied keywords (`search.list`).
- Fetch detailed video statistics such as view count, like count, and comment count (`videos.list`).
- Retrieve channel-level information including subscriber count and channel description (`channels.list`).
- Obtain the mapping of video category IDs to human-readable category names (`videoCategories.list`).

Data Use Limitation: All data retrieved from the YouTube Data API v3 is used solely to display analytics information within the Verdent application. No data is sold, redistributed, or used for advertising purposes.

2. API Usage Flow

The following steps describe the complete data-retrieval flow triggered when a user performs a keyword search in Verdent. Each step maps to one or more YouTube Data API v3 endpoints.

Step 1

User Enters a Keyword → `search.list`

The user types a keyword (e.g., "AI tutorial") and optionally selects a date-range filter (past 24 h, past week, past month, or custom). Verdent calls `search.list` with parameters: `part=snippet`, `type=video`, `maxResults=50`, and `publishedAfter/publishedBefore` if a date filter is active. The response returns a list of video IDs and snippet metadata (title, channel ID, thumbnail URL, publish date).

Step 2

Video IDs Collected → `videos.list`

The video IDs returned by `search.list` are batched (up to 50 per request) and passed to `videos.list` with `part=statistics,snippet,contentDetails`. This retrieves `viewCount`, `likeCount`, `commentCount`, `duration`, and category ID for each video. Batching minimises quota consumption.

Step 3

Channel IDs Collected → channels.list

Unique channel IDs extracted from the search results are batched and sent to **channels.list** with *part=statistics,snippet*. This provides subscriberCount, videoCount, viewCount, and the channel thumbnail URL displayed alongside each result.

Step 4

Results Displayed with Estimated Revenue

All retrieved data is merged in-memory and rendered in the Verdent UI. An estimated revenue figure is derived from publicly available CPM benchmarks and the video's viewCount: *Estimated Revenue = CPM x viewCount / 1000*. This figure is labelled as an estimate and is never returned by the API.

Quota Optimisation Strategy

Verdent implements two layers of quota conservation to comply with the YouTube API Services usage policies:

Strategy	Details	Quota Saving
Daily Search Cache	Results for identical keyword + date-range combinations are cached for 24 hours and listed as a single API call.	Each day for 24 hours and listed as a single API call.
Batch API Requests	video IDs and channel IDs are grouped into batches of up to 50, reducing the number of API calls by up to 50%.	Reducing the number of API calls by up to 50%.
On-Demand Only	API calls are triggered only by explicit user action; no background or periodic automatic refresh.	Zero idle quota usage.

3. Sample API Response Data

This section provides annotated excerpts of actual API responses returned by the YouTube Data API v3. Sensitive personal data (OAuth tokens, internal user IDs) is not involved because the application uses API-key-only authentication for public data.

3.1 search.list Response Excerpt

The following is a representative JSON excerpt returned by **search.list** for the query "AI tutorial" with **publishedAfter** set to 7 days prior to the request date:

```
{
  "kind": "youtube#searchListResponse",
  "etag": "someEtagValue",
  "nextPageToken": "CAUQAA",
  "regionCode": "JP",
  "pageInfo": { "totalResults": 1000000, "resultsPerPage": 50 },
  "items": [
    {
      "kind": "youtube#searchResult",
      "id": { "kind": "youtube#video", "videoId": "dQw4w9WgXcQ" },
      "snippet": {
        "publishedAt": "2026-06-14T10:30:00Z",
        "channelId": "UCxxxxxxxxxxxxxxxxxxxxxx",
        "title": "Complete AI Tutorial for Beginners 2026",
        "description": "Learn AI from scratch ...",
        "channelTitle": "TechLearn Pro",
        "thumbnails": {
          "medium": {
            "url": "https://i.ytimg.com/vi/dQw4w9WgXcQ/mqdefault.jpg",
            "width": 320, "height": 180
          }
        },
        "liveBroadcastContent": "none"
      }
    },
    // ... up to 50 items per page
  ]
}
```

Field	How Verdent uses it
videoId	Used as the key for subsequent videos.list calls (batched).
channelId	Collected for a batched channels.list call.
title / description	Displayed verbatim in the search results UI.
thumbnails.medium.url	Displayed as a thumbnail image; the URL is not stored persistently.
publishedAt	Used to calculate relative post age (e.g., '5 days ago') shown in the UI.

3.2 videos.list Response Excerpt

```
{
  "kind": "youtube#videoListResponse",
  "items": [
    {
      "kind": "youtube#video",
      "id": "dQw4w9WgXcQ",
      "snippet": { "categoryId": "28" },
      "statistics": {
        "viewCount": "1523847",
        "likeCount": "48201",
        "commentCount": "3912"
      },
      "contentDetails": { "duration": "PT14M32S" }
    }
  ]
}
```

3.3 Caching Strategy

Search results are cached in a local SQLite database using the following schema. The cache key is a SHA-256 hash of the query parameters (keyword + date range). Cache entries expire after 86,400 seconds (24 hours), after which the next request triggers a fresh API call.

```
CREATE TABLE search_cache (
  cache_key TEXT PRIMARY KEY, -- SHA-256(keyword + date_range)
  response TEXT NOT NULL, -- serialised JSON response
  fetched_at INTEGER NOT NULL, -- Unix timestamp
  expires_at INTEGER NOT NULL -- fetched_at + 86400
);
```

4. Information Displayed to Users

The following table enumerates every data field originating from the YouTube Data API v3 that is rendered in the Verdent user interface, together with the API source and the display context.

Field	API Source	Display Context	Stored?
Video thumbnail	search.list (snippet.thumbnails)	Search result card	No (URL only, in cache)
Video title	search.list (snippet.title)	Search result card, detail panel	Yes (in cache)
Channel name	search.list (snippet.channelTitle)	Search result card	Yes (in cache)
Publish date	search.list (snippet.publishedAt)	Relative age label	Yes (in cache)
View count	videos.list (statistics.viewCount)	Metric card, trend chart	Yes (in cache)
Like count	videos.list (statistics.likeCount)	Engagement rate calc.	Yes (in cache)
Comment count	videos.list (statistics.commentCount)	Engagement rate calc.	Yes (in cache)
Video duration	videos.list (contentDetails.duration)	Duration badge	Yes (in cache)
Subscriber count	channels.list (statistics.subscriberCount)	Channel info panel	Yes (in cache)
Channel thumbnail	channels.list (snippet.thumbnails)	Channel info panel	No (URL only, in cache)
Video category	videoCategories.list	Category filter tag	Yes (look-up table)

4.1 Derived Metrics (Not Returned by the API)

Verdent computes two derived metrics from the raw API data. These values are clearly labelled as estimates in the UI.

Metric	Formula	Label in UI
Engagement Rate	$(\text{likeCount} + \text{commentCount}) / \text{viewCount} \times 100$	Engagement Rate (%)
Estimated Revenue	$\text{CPM (user-configurable, default \$2)} \times \text{viewCount} / 1000$	Est. Revenue (USD) — Estimate Only

4.2 Trend Analytics

Verdent aggregates the video data retrieved for a given search to render the following trend charts. All chart data is derived exclusively from API fields listed in the table above:

- Upload time-of-day distribution (bar chart) — derived from publishedAt.
- Day-of-week distribution (bar chart) — derived from publishedAt.
- View count distribution (histogram) — derived from statistics.viewCount.
- Subscriber count vs. view count scatter plot — channel-level correlation.
- Category breakdown (pie chart) — derived from videoCategories.list.

No Personal Data: None of the fields displayed or computed by Verdent constitute personal data as defined under applicable privacy regulations. Verdent does not collect, store, or process any information that could identify individual YouTube viewers.

5. YouTube API Services Terms of Service Compliance

Verdent has reviewed and acknowledges all provisions of the **YouTube API Services Terms of Service** (<https://developers.google.com/youtube/terms/api-services-terms-of-service>) and the **YouTube API Services Developer Policies**. The following sub-sections describe how the application satisfies each material requirement.

5.1 Acceptable Use

Read-Only Access

All API calls use GET requests only. Verdent does not create, update, or delete YouTube resources. No OAuth scopes beyond public-data access are requested.

No Data Scraping

Verdent relies solely on the YouTube Data API v3 for data retrieval. No HTML scraping, reverse-engineered endpoints, or unofficial YouTube interfaces are used.

No Unauthorised Redistribution

API data is displayed only within the Verdent application interface. No bulk data export, resale, or redistribution to third parties occurs.

Attribution

All video and channel data is attributed to YouTube / Google LLC in the application UI footer and in the application's About page, as required by the Branding Guidelines.

Quota Compliance

The application monitors its daily quota consumption and displays a warning when usage exceeds 80% of the allocated quota. Automated back-off logic prevents quota exhaustion.

5.2 User Privacy and Data Protection

Verdent does not authenticate end-users via Google/YouTube OAuth. No user viewing history, liked videos, subscriptions, or other personally identifiable information is requested or stored. The application's privacy policy (<https://verdent.app/privacy>) explicitly states:

- We do not collect YouTube user account data.
- We do not store API responses beyond the 24-hour cache window.
- Cached data is stored locally on the user's device and is never transmitted to Verdent servers.
- Users may clear the local cache at any time via the application Settings.

5.3 Prohibited Use Confirmation

Verdent confirms that it does NOT engage in any of the following activities prohibited by the YouTube API Services Terms of Service:

- | | |
|--|--------------------|
| • Using API data to build a competing video-hosting or streaming platform. | Confirmed Not Done |
|--|--------------------|

• Bypassing or circumventing YouTube's advertising systems.	Confirmed Not Done
• Using the API to facilitate spam, fraud, or manipulation of engagement metrics.	Confirmed Not Done
• Storing API data for longer than permitted under the data deletion requirements.	Confirmed Not Done
• Accessing non-public data without appropriate OAuth authorisation.	Confirmed Not Done
• Embedding YouTube content outside of the YouTube player in violation of policy.	Confirmed Not Done

Developer Acknowledgement

The developer and operator of Verdent acknowledges that the application has been built and is operated in full compliance with the YouTube API Services Terms of Service, the YouTube API Services Developer Policies, and Google's Privacy Policy. The developer agrees to promptly address any compliance concerns raised by Google and to update the application as required to maintain compliance with future policy changes.

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Application URL:	https://verdent.app
Report Date:	June 2026
Signature:	_____

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